AI & ML Timeline

From Turing’s question to the generative AI race

Machine Learning Fundamentals | 1.3 Discussion: AI & ML Timelines | Aashirwad Shrestha



1950s
Foundations



1974–93
AI winters



1980s–2000s
Boom & revival

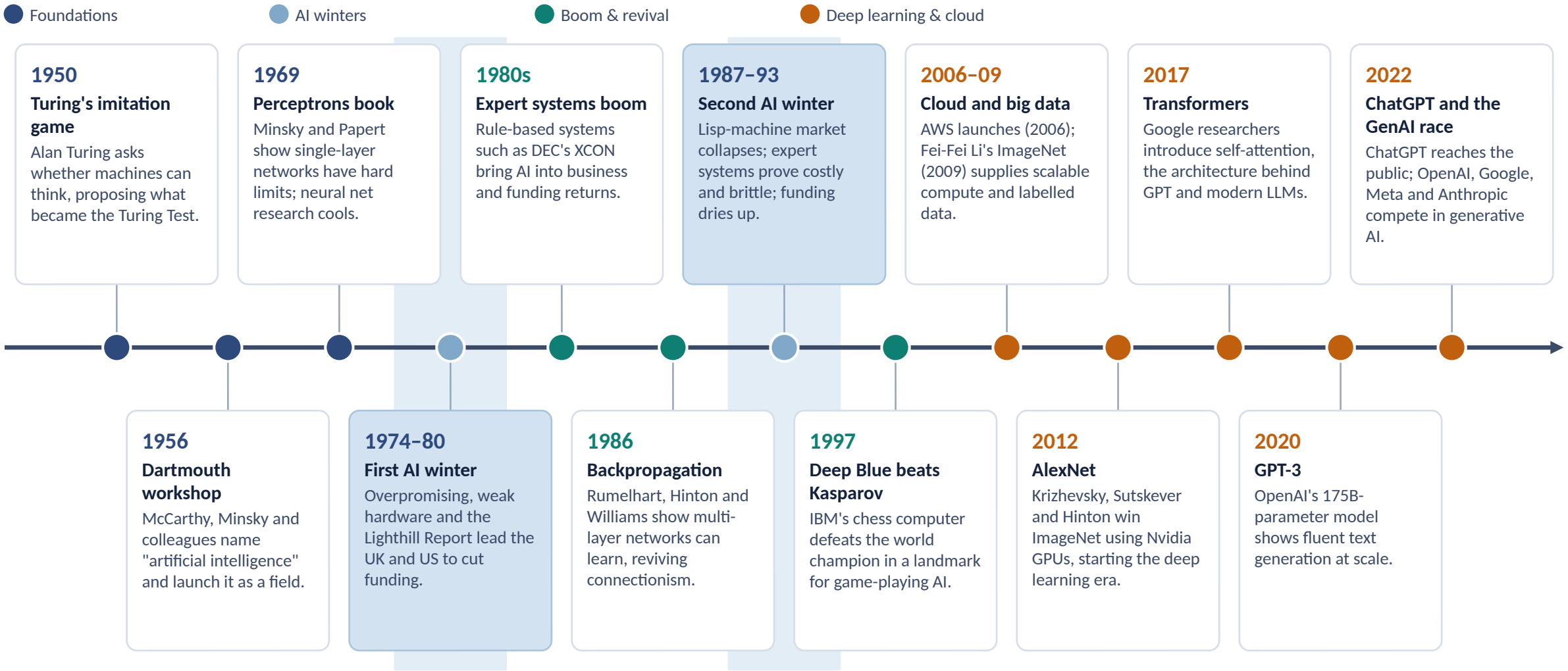


2006–now
Deep learning & cloud

Roadmap: Milestones | AI winters | Deep learning and cloud | Leading models and companies | Industry impact

The Big Picture, 1950–2022

Training compute doubling time
~20 months until ~2010, ~6 months since (Roser, 2022)



Sources: Putchuon (2024); Bhat (2025); Roser (2022); primary sources on the References slide.

Foundations: 1943–1969

1943

First artificial neuron

McCulloch and Pitts publish a mathematical model of a neuron, the seed of neural networks.

1950

The Turing Test

Alan Turing’s paper "Computing Machinery and Intelligence" proposes judging machine intelligence by indistinguishable behavior.

1956

Dartmouth Conference

Often called the birthplace of AI as a field. John McCarthy coined the term; Minsky, Shannon and Rochester co-organized.

1958

The perceptron

Frank Rosenblatt builds a learning machine that adjusts weights from examples, an early ancestor of modern ML.

1966

ELIZA

Joseph Weizenbaum’s chatbot at MIT mimics a therapist with simple pattern rules, an early natural-language demo.

1969

Perceptrons critique

Minsky and Papert prove single-layer perceptrons cannot learn problems like XOR, shifting attention to symbolic AI.

Era theme: Symbolic AI. Intelligence was seen as logic and rules, with big promises and very little computing power.

Sources: Bhat (2025); McCulloch & Pitts (1943); Turing (1950); McCarthy et al. (1955); Rosenblatt (1958); Weizenbaum (1966); Minsky & Papert (1969).

The AI Winters: Causes and Impacts

First AI Winter (c. 1974–1980)

Causes

- Early promises (machine translation, general problem solving) went unmet
- Computers had far too little speed and memory
- Perceptron limits (1969) stalled neural nets; the 1973 Lighthill Report judged AI progress poor

Impacts

- UK government cut funding at most universities
- US agencies such as DARPA tightened funding toward results-focused projects
- Researchers left the field or rebranded their work

Second AI Winter (c. 1987–1993)

Causes

- Cheap desktop workstations made specialized Lisp machines obsolete
- Expert systems were costly to maintain, brittle, and could not learn
- Ambitious government programs, such as Japan's Fifth Generation project, fell short

Impacts

- Lisp-machine makers and AI start-ups collapsed
- Funding dried up again; "AI" became a label to avoid
- Work continued quietly as machine learning, statistics and robotics

Pattern: hype, then limits, then funding cuts. Each thaw came from new methods plus more computing power.

Sources: Bhat (2025); Lighthill (1973); Crevier (1993); Russell & Norvig (2020).

Revival: 1980s–2011

1980

XCON expert system

DEC's rule-based XCON configured computer orders and was reported to save the company millions, sparking the expert-system boom.

1986

Backpropagation

Rumelhart, Hinton and Williams popularize a method for training multi-layer neural networks.

1997

Deep Blue

IBM's chess machine beats world champion Garry Kasparov using massive search on custom hardware.

2006

Deep belief networks

Hinton, Osindero and Teh show deep networks can be trained layer by layer, relaunching "deep learning".

2011

IBM Watson

Watson defeats human champions on Jeopardy!, showing question answering over natural language at scale.

Takeaway: Progress came from narrow, practical systems and better learning algorithms, not from general reasoning.

Sources: McDermott (1982); Rumelhart et al. (1986); Campbell et al. (2002); Hinton et al. (2006); Ferrucci et al. (2010); Bhat (2025).

Deep Learning and Cloud: 2006–2017

Compute: GPUs

Nvidia's CUDA (2007) let GPUs run general computation. Jensen Huang's company became the hardware backbone of deep learning.

Cloud: rentable scale

Amazon Web Services (2006), then Google Cloud and Microsoft Azure, let researchers rent huge compute without owning it.

Data: ImageNet

Fei-Fei Li's team released ImageNet (2009), millions of labelled images that made large-scale training possible.

2012

AlexNet

Krizhevsky, Sutskever and Hinton cut ImageNet error sharply using GPUs; about 470 petaFLOP of training.

2014

GANs

Ian Goodfellow introduces generative adversarial networks, a key step to AI-made images.

2016

AlphaGo

DeepMind's AlphaGo beats Go champion Lee Sedol 4–1, a feat once thought a decade away.

2017

Transformer

"Attention Is All You Need" (Google) introduces self-attention, enabling today’s large language models.

Sources: Roser (2022); Putchuon (2024); Krizhevsky et al. (2012); Goodfellow et al. (2014); Silver et al. (2016); Vaswani et al. (2017); Deng et al. (2009).

Leading AI Companies and Models

Company	Key models and products	Main contribution
OpenAI	GPT-3, ChatGPT, GPT-4, DALL-E	Brought large language models to the public with ChatGPT (2022)
Google / DeepMind	Transformer, BERT, AlphaGo, AlphaFold, PaLM, Gemini	Core research breakthroughs, from game-playing to protein structure
Anthropic	Claude	Safety-focused large language models, founded 2021 by former OpenAI researchers
Meta	Llama	Widely used open-weight models that broadened access to LLMs
Microsoft	Copilot, Azure AI	Major OpenAI partner; cloud platform for enterprise AI
Nvidia	GPUs, CUDA	Hardware that made deep learning practical (the AlexNet moment)

Sources: Putchuon (2024); Bhat (2025); Devlin et al. (2019); Brown et al. (2020); Jumper et al. (2021); Touvron et al. (2023); OpenAI (2022). Model lineups change quickly.

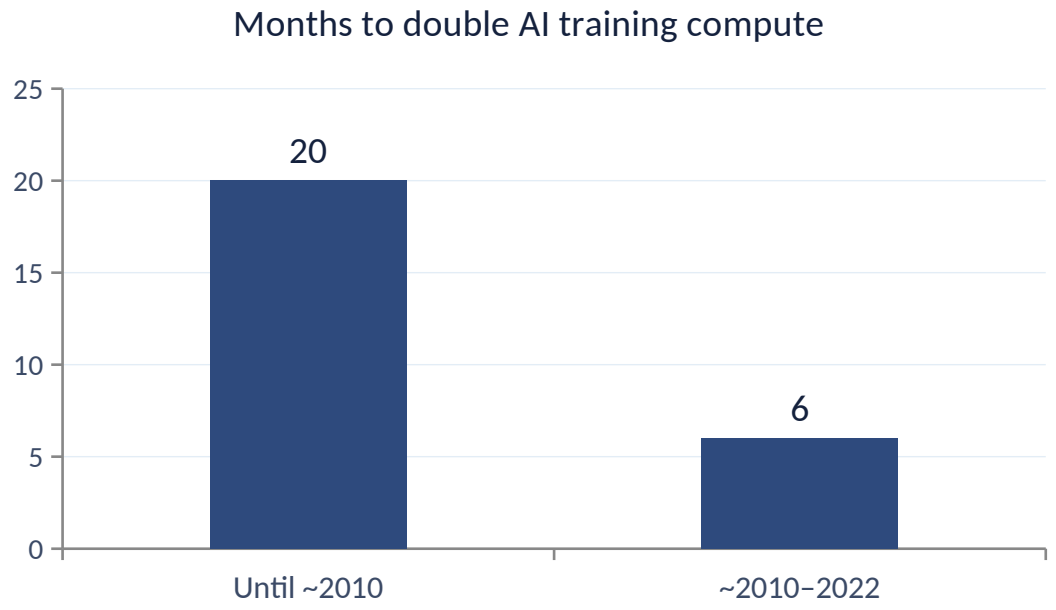
Key Concepts and the Compute Surge

Symbolic AI vs Connectionism

Symbolic AI (1950s–80s) writes knowledge as hand-coded rules and logic, as in expert systems. Connectionism uses neural networks that learn patterns from data and drives today's deep learning.

Narrow AI vs AGI

Narrow AI does specific tasks such as chess, image recognition or translation; every system built so far is narrow. AGI would match humans across tasks and does not exist yet; expert forecasts vary widely.



Doubling now takes about a third of the time. PaLM (2022) used over 5 million times the training compute of AlexNet (2012).

Sources: Russell & Norvig (2020); Roser (2022), data from Sevilla et al. (2022).

Impact Across Industries

Healthcare

Algorithms read medical images to support diagnosis; AI models simulate chemistry to speed drug discovery.

Finance

Models assess risk, detect fraud and, as Roser notes, increasingly help decide who gets a loan.

Transport

Self-driving systems process live sensor data; AI also sets airline prices and assists pilots.

Retail and media

Recommender systems at Netflix, Amazon and YouTube shape what billions of people watch and buy.

Creative industries

Generative AI drafts text, images, music and scripts, raising new questions about authorship and deepfakes.

Software and education

AI helps write code and translate text, and personalizes learning materials for each student.

AI is dual-use: the same technology can drive real benefits and real harms, so understanding it matters for everyone.

Sources: Bhat (2025); Roser (2022).

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AI use: I used Claude (Anthropic) to help research and draft this timeline.